

UNIT 3 – LAB EXPERIMENTS (WEKA)

CSA1610 – DWDM

G.Hima Bindu(192511377)

EXPERIMENT 1:

Consider the data set and perform the Apriori Algorithm and FP algorithm support:3 and confidence=50%

Customer ID	Transaction ID	Items Bought
1	0001	{a, d, e}
1	0024	{a, b, c, e}
2	0012	{a, b, d, e}
2	0031	{a, c, d, e}
3	0015	{b, c, e}
3	0022	{b, d, e}
4	0029	{c, d}
4	0040	{a, b, c}
5	0033	{a, d, e}
5	0038	{a, b, e}

Step 1: Prepare the Dataset

You'll need to convert the table into **ARFF format** (WEKA's native file type). Here's how your .arff file should look:

arff

@relation transactions

@attribute a {t,f}

@attribute b {t,f}

@attribute c {t,f}

@attribute d {t,f}

@attribute e {t,f}

@data

t,f,f,t,t

t,t,t,f,t

t,t,f,t,t

t,f,t,t,t

f,t,t,f,t

f,t,f,t,t

f,f,t,t,f

t,t,t,f,f

t,f,f,t,t

t,t,f,f,t

Note:

- Each transaction is represented as a set of items.
- You can save this file as transactions.arff.

The screenshot shows the Weka Explorer interface with the 'Associate' tab selected. The 'Apriori' algorithm is chosen with the following command: `Apriori -N 10 -T 0 -C 0.5 -D 0.05 -U 1.0 -M 0.3 -S -1.0 -c -1`. The 'Associator output' pane displays the results of the mining process.

Result list (right-click for ...): 21:43:04 - Apriori

Associator output:

```
==== Associator model (full training set) ====

Apriori
=====

Minimum support: 0.45 (4 instances)
Minimum metric <confidence>: 0.5
Number of cycles performed: 11

Generated sets of large itemsets:

Size of set of large itemsets L(1): 8

Size of set of large itemsets L(2): 10

Size of set of large itemsets L(3): 3

Best rules found:

1. c=f 5 ==> e=t 5    <conf:(1)> lift:(1.25) lev:(0.1) [0] conv:(1)
2. d=f 4 ==> b=t 4    <conf:(1)> lift:(1.67) lev:(0.16) [1] conv:(1.6)
3. b=f 4 ==> d=t 4    <conf:(1)> lift:(1.67) lev:(0.16) [1] conv:(1.6)
4. a=t c=f 4 ==> e=t 4    <conf:(1)> lift:(1.25) lev:(0.08) [0] conv:(0.8)
5. a=t d=t 4 ==> e=t 4    <conf:(1)> lift:(1.25) lev:(0.08) [0] conv:(0.8)
6. c=f d=t 4 ==> e=t 4    <conf:(1)> lift:(1.25) lev:(0.08) [0] conv:(0.8)
7. a=t 7 ==> e=t 6    <conf:(0.86)> lift:(1.07) lev:(0.04) [0] conv:(0.7)
8. b=t 6 ==> e=t 5    <conf:(0.83)> lift:(1.04) lev:(0.02) [0] conv:(0.6)
9. d=t 6 ==> e=t 5    <conf:(0.83)> lift:(1.04) lev:(0.02) [0] conv:(0.6)
10. c=f 5 ==> a=t 4    <conf:(0.8)> lift:(1.14) lev:(0.05) [0] conv:(0.75)
```

Status: OK Log x 0

EXPERIMENT 2:

Consider the data set and perform the Apriori Algorithm and FP algorithm support:3 and confidence=50%

Consider the market basket transactions shown in the above table.

(a) What is the maximum number of association rules that can be extracted from this data (including rules that have zero support)?

(b) What is the maximum size of frequent itemsets that can be extracted (assuming minsup > 0)?

Transaction ID	Items Bought
1	{Milk, Beer, Diapers}
2	{Bread, Butter, Milk}
3	{Milk, Diapers, Cookies}
4	{Bread, Butter, Cookies}
5	{Beer, Cookies, Diapers}
6	{Milk, Diapers, Bread, Butter}
7	{Bread, Butter, Diapers}
8	{Beer, Diapers}
9	{Milk, Diapers, Bread, Butter}
10	{Beer, Cookies}

Step 1: Create the ARFF file

You have six items: **Milk, Beer, Diapers, Bread, Butter, Cookies**. Represent each item as a Boolean attribute (t = present, f = absent).

arff

@relation market_basket

@attribute Milk {t,f}

@attribute Beer {t,f}

@attribute Diapers {t,f}

@attribute Bread {t,f}

@attribute Butter {t,f}

@attribute Cookies {t,f}

@data

t,f,t,f,f,f

f,t,t,t,t,f

t,f,t,f,f,t

f,t,f,f,f,t

t,t,f,f,f,t

f,t,t,t,t,f

t,t,t,f,f,f

f,t,t,f,f,f

t,t,t,t,t,f

f,t,f,f,f,t

Save this file as market_basket.arff.

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Associate

Choose Apriori -N 10 -T 0 -C 0.5 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1

Start Stop

Result list (right-click for ...)

220050 - Apriori

Apriori output

Run information ==

Scheme: weka.associations.Apriori -N 10 -T 0 -C 0.5 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1

Relation: market_basket

Instances: 10

Attributes: 6

Milk

Beer

Diapers

Bread

Butter

Cookies

== Apriori model (full training set) ==

Apriori

=====

Minimum support: 0.50 (5 instances)

Minimum metric (confidence): 0.5

Number of cycles performed: 9

Generated sets of large itemsets:

Size of set of large itemsets L(1): 7

Size of set of large itemsets L(2): 7

Size of set of large itemsets L(3): 2

Best rules found:

1. Butter=f 7 ==> Bread=f 7 <conf:(1)> lift:(1.43) lev:(0.21) [2] conv:(2.1)
2. Bread=f 7 ==> Butter=f 7 <conf:(1)> lift:(1.43) lev:(0.21) [2] conv:(2.1)
3. Cookies=f 6 ==> Diapers=t 6 <conf:(1)> lift:(1.43) lev:(0.18) [1] conv:(1.8)
4. Milk=f 5 ==> Beer=t 5 <conf:(1)> lift:(1.25) lev:(0.1) [0] conv:(1)
5. Beer=t Cookies=f 5 ==> Diapers=t 5 <conf:(1)> lift:(1.43) lev:(0.15) [1] conv:(1.5)
6. Beer=t Diapers=t 5 ==> Cookies=f 5 <conf:(1)> lift:(1.67) lev:(0.2) [2] conv:(2)
7. Beer=t Butter=f 5 ==> Bread=f 5 <conf:(1)> lift:(1.43) lev:(0.15) [1] conv:(1.5)
8. Beer=t Bread=f 5 ==> Butter=f 5 <conf:(1)> lift:(1.43) lev:(0.15) [1] conv:(1.5)
9. Diapers=t 7 ==> Cookies=f 6 <conf:(0.86)> lift:(1.43) lev:(0.18) [1] conv:(1.4)
10. Cookies=f 6 ==> Beer=t 5 <conf:(0.83)> lift:(1.04) lev:(0.02) [0] conv:(0.6)

Status: OK

Log

Best rules found:

1. Butter=f 7 ==> Bread=f 7 <conf:(1)> lift:(1.43) lev:(0.21) [2] conv:(2.1)
2. Bread=f 7 ==> Butter=f 7 <conf:(1)> lift:(1.43) lev:(0.21) [2] conv:(2.1)
3. Cookies=f 6 ==> Diapers=t 6 <conf:(1)> lift:(1.43) lev:(0.18) [1] conv:(1.8)
4. Milk=f 5 ==> Beer=t 5 <conf:(1)> lift:(1.25) lev:(0.1) [0] conv:(1)
5. Beer=t Cookies=f 5 ==> Diapers=t 5 <conf:(1)> lift:(1.43) lev:(0.15) [1] conv:(1.5)
6. Beer=t Diapers=t 5 ==> Cookies=f 5 <conf:(1)> lift:(1.67) lev:(0.2) [2] conv:(2)
7. Beer=t Butter=f 5 ==> Bread=f 5 <conf:(1)> lift:(1.43) lev:(0.15) [1] conv:(1.5)
8. Beer=t Bread=f 5 ==> Butter=f 5 <conf:(1)> lift:(1.43) lev:(0.15) [1] conv:(1.5)
9. Diapers=t 7 ==> Cookies=f 6 <conf:(0.86)> lift:(1.43) lev:(0.18) [1] conv:(1.4)
10. Cookies=f 6 ==> Beer=t 5 <conf:(0.83)> lift:(1.04) lev:(0.02) [0] conv:(0.6)

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Associator

Choose **FPGrowth** -P 2 -I -1 -N 10 -T 0 -C 0.5 -D 0.05 -U 1.0 -M 0.1

Start

Stop

Result list (right-click for ...)

22:00:50 - Apriori

22:04:55 - FPGrowth

Associator output

```
=== Run information ===

Scheme:      weka.associations.FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.5 -D 0.05 -U 1.0 -M 0.1
Relation:     market_basket
Instances:    10
Attributes:   6
              Milk
              Beer
              Diapers
              Bread
              Butter
              Cookies

=== Associator model (full training set) ===

FPGrowth found 19 rules (displaying top 10)

1. [Butter=f]: 7 ==> [Bread=f]: 7   <conf:(1)> lift:(1.43) lev:(0.21) conv:(2.1)
2. [Bread=f]: 7 ==> [Butter=f]: 7   <conf:(1)> lift:(1.43) lev:(0.21) conv:(2.1)
3. [Diapers=f]: 3 ==> [Butter=f]: 3   <conf:(1)> lift:(1.43) lev:(0.09) conv:(0.9)
4. [Diapers=f]: 3 ==> [Bread=f]: 3   <conf:(1)> lift:(1.43) lev:(0.09) conv:(0.9)
5. [Butter=f, Cookies=f]: 3 ==> [Bread=f]: 3   <conf:(1)> lift:(1.43) lev:(0.09) conv:(0.9)
6. [Bread=f, Cookies=f]: 3 ==> [Butter=f]: 3   <conf:(1)> lift:(1.43) lev:(0.09) conv:(0.9)
7. [Butter=f, Milk=f]: 3 ==> [Bread=f]: 3   <conf:(1)> lift:(1.43) lev:(0.09) conv:(0.9)
8. [Bread=f, Milk=f]: 3 ==> [Butter=f]: 3   <conf:(1)> lift:(1.43) lev:(0.09) conv:(0.9)
9. [Diapers=f]: 3 ==> [Butter=f, Bread=f]: 3   <conf:(1)> lift:(1.43) lev:(0.09) conv:(0.9)
10. [Butter=f, Diapers=f]: 3 ==> [Bread=f]: 3   <conf:(1)> lift:(1.43) lev:(0.09) conv:(0.9)
```

EXPERIMENT 3:

Bayes classification and descion tree (using training and test data)

R/D	age	income	student	credit_rating	Class: buys_computer
1	<=30	high	no	fair	no
2	<=30	high	no	excellent	no
3	31 ... 40	high	no	fair	yes
4	>40	medium	no	fair	yes
5	>40	low	yes	fair	yes
6	>40	low	yes	excellent	no
7	31 ... 40	low	yes	excellent	yes
8	<=30	medium	no	fair	no
9	<=30	low	yes	fair	yes
10	>40	medium	yes	fair	yes
11	<=30	medium	yes	excellent	yes
12	31 ... 40	medium	no	excellent	yes
13	31 ... 40	high	yes	fair	yes
14	>40	medium	no	excellent	no

Step 1: Create the ARFF file

Save the following as buys_computer.arff:

```
arff
```

```
@relation buys_computer
```

```
@attribute age {<=30,31-40,>40}
```

```
@attribute income {high,medium,low}
```

```
@attribute student {yes,no}
```

```
@attribute credit_rating {fair,excellent}
```

```
@attribute buys_computer {yes,no}
```

```
@data
```

```
<=30,high,no,fair,no
```

```
<=30,high,no,excellent,no
```

```
31-40,high,no,fair,yes
```

```
>40,medium,no,fair,yes
```

```
>40,low,yes,fair,yes
```

>40,low,yes,excellent,no

31-40,low,yes,excellent,yes

<=30,medium,no,fair,no

<=30,low,yes,fair,yes

>40,medium,yes,fair,yes

<=30,medium,yes,excellent,yes

31-40,medium,no,excellent,yes

31-40,high,yes,fair,yes

>40,medium,no,excellent,no

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose NaiveBayes

Test options

☐ Use training set

☐ Supplied test set Set...

☒ Cross-validation Folds 10

☐ Percentage split % 66

More options...

(Nom) buys_computer

Start Stop

Result list (right-click for options)

22:09:46 - bayes.NaiveBayes

Classifier output

```
fair      7.0    3.0
excellent  4.0    4.0
[total]   11.0   7.0
```

Time taken to build model: 0 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	8	57.1429 %
Incorrectly Classified Instances	6	42.8571 %
Kappa statistic	-0.0244	
Mean absolute error	0.4374	
Root mean squared error	0.4916	
Relative absolute error	91.8631 %	
Root relative squared error	99.6492 %	
Total Number of Instances	14	

=== Detailed Accuracy By Class ===

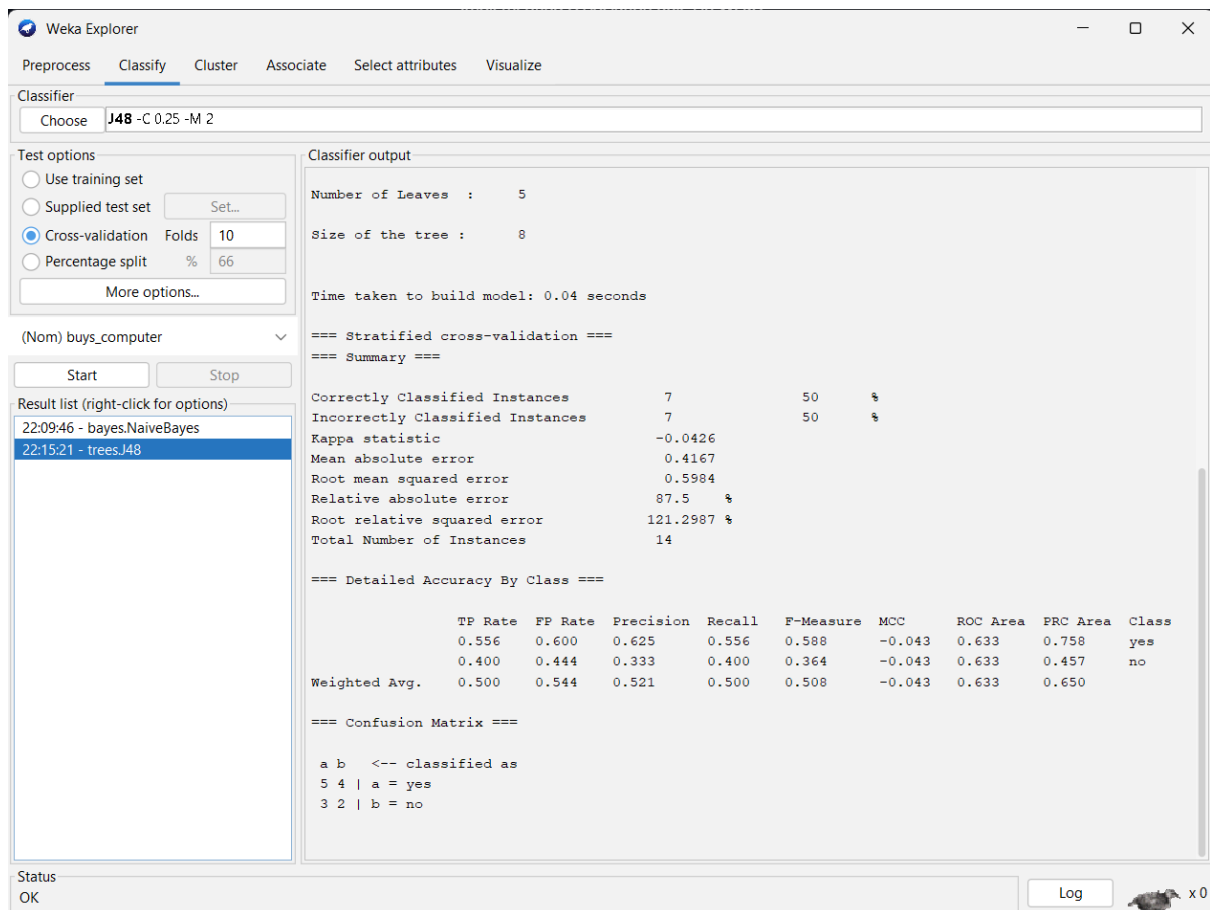
	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.778	0.800	0.636	0.778	0.700	-0.026	0.578	0.697	yes
	0.200	0.222	0.333	0.200	0.250	-0.026	0.578	0.557	no
Weighted Avg.	0.571	0.594	0.528	0.571	0.539	-0.026	0.578	0.647	

=== Confusion Matrix ===

```
a b  <-- classified as
7 2 | a = yes
4 1 | b = no
```

Status OK

Log x 0



EXPERIMENT 4:

Analysis the dataset “diabetes.csv” how the diabetes trend is for different age people, using linear regression and multiple regression.

ARFF CODE:

@relation diabetes_trend

@attribute Age numeric

@attribute BMI numeric

@attribute Glucose numeric

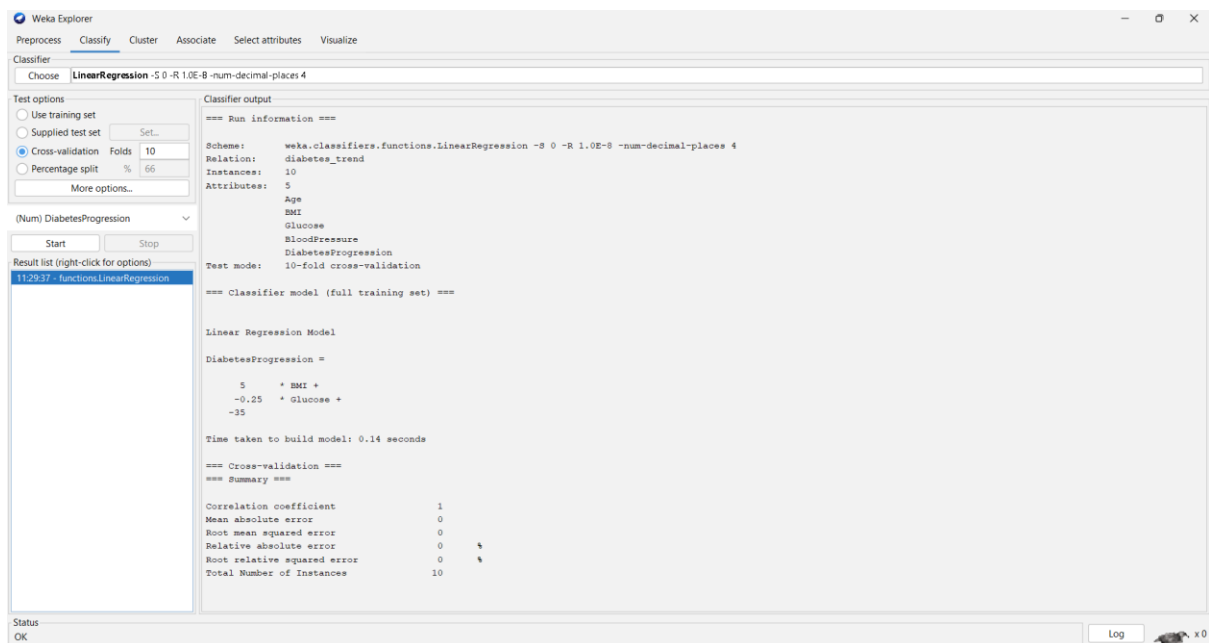
@attribute BloodPressure numeric

@attribute DiabetesProgression numeric

@data

25,22.5,130,80,45

30,24.0,140,85,50
 35,26.5,150,88,60
 40,28.0,160,90,65
 45,29.5,170,92,70
 50,31.0,180,95,75
 55,32.5,190,98,80
 60,34.0,200,100,85
 65,35.5,210,102,90
 70,37.0,220,105,95



EXPERIMENT 5:

Implement using WEKA for the given Suppose a database has five transactions. Let min sup= 50%(2) and min con f = 80%.

Transactions	Items
T1	(M, O, N, K, E, Y)
T2	(D, O, N, K, E, Y)
T3	(M, A, K, E)
T4	(M, U, C, K, Y)
T5	(C, O, O, K, I, E)

- Find all frequent item sets using Apriori algorithm
- Also draw FP-Growth Tree

ARFF CODE:

@relation market_basket

@attribute M {t,f}

@attribute O {t,f}

@attribute N {t,f}

@attribute K {t,f}

@attribute E {t,f}

@attribute Y {t,f}

@attribute D {t,f}

@attribute A {t,f}

@attribute U {t,f}

@attribute C {t,f}

@attribute I {t,f}

@data

t,t,t,t,t,f,f,f,f

f,t,t,t,t,t,f,f,f

t,f,f,t,t,f,t,f,f

t,f,f,t,t,f,t,f,f

f,t,f,t,t,f,f,f,t

Weka Explorer

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator

Choose **Apriori** -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1

Start Stop

Result list (right-click for ...)

11:36:08 - Apriori

Associator output

```

C
I
=== Associator model (full training set) ===

Apriori
=====

Minimum support: 0.85 (4 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 3

Generated sets of large itemsets:

Size of set of large itemsets L(1): 7

Size of set of large itemsets L(2): 8

Size of set of large itemsets L(3): 2

Best rules found:

1. E=t 4 ==> K=t 4 <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
2. D=f 4 ==> K=t 4 <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
3. A=f 4 ==> K=t 4 <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
4. U=f 4 ==> K=t 4 <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
5. C=f 4 ==> K=t 4 <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
6. I=f 4 ==> K=t 4 <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
7. U=f 4 ==> E=t 4 <conf:(1)> lift:(1.25) lev:(0.16) [0] conv:(0.8)
8. E=t 4 ==> U=f 4 <conf:(1)> lift:(1.25) lev:(0.16) [0] conv:(0.8)
9. I=f 4 ==> C=f 4 <conf:(1)> lift:(1.25) lev:(0.16) [0] conv:(0.8)
10. C=f 4 ==> I=f 4 <conf:(1)> lift:(1.25) lev:(0.16) [0] conv:(0.8)

```

Status OK Log x 0

Weka Explorer

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator

Choose **FPGrowth** -P 2 -I -1 -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1

Start Stop

Result list Starts the associator

11:36:08 - Apriori

11:36:25 - FPGrowth

Associator output

```

=== Run information ===

Scheme:      weka.associations.FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1
Relation:    market_basket
Instances:   5
Attributes:  11
M
O
N
K
E
Y
D
A
U
C
I

=== Associator model (full training set) ===

FPGrowth found 133 rules (displaying top 10)

1. [Y=f]: 2 ==> [U=f]: 2 <conf:(1)> lift:(1.25) lev:(0.08) conv:(0.4)
2. [M=f]: 2 ==> [U=f]: 2 <conf:(1)> lift:(1.25) lev:(0.08) conv:(0.4)
3. [I=f]: 4 ==> [C=f]: 4 <conf:(1)> lift:(1.25) lev:(0.16) conv:(0.8)
4. [C=f]: 4 ==> [I=f]: 4 <conf:(1)> lift:(1.25) lev:(0.16) conv:(0.8)
5. [O=f]: 2 ==> [I=f]: 2 <conf:(1)> lift:(1.25) lev:(0.08) conv:(0.4)
6. [N=f]: 3 ==> [D=f]: 3 <conf:(1)> lift:(1.25) lev:(0.12) conv:(0.6)
7. [Y=f]: 2 ==> [D=f]: 2 <conf:(1)> lift:(1.25) lev:(0.08) conv:(0.4)
8. [O=f]: 2 ==> [D=f]: 2 <conf:(1)> lift:(1.25) lev:(0.08) conv:(0.4)
9. [O=f]: 2 ==> [C=f]: 2 <conf:(1)> lift:(1.25) lev:(0.08) conv:(0.4)
10. [M=f]: 2 ==> [A=f]: 2 <conf:(1)> lift:(1.25) lev:(0.08) conv:(0.4)

```

Status OK Log x 0

EXPERIMENT 6:

Prediction of Categorical Data using Decision Tree Algorithm through WEKA using any datasets. a) Tree b) Preprocess c) Logistic

ARFF CODE:

@relation product_purchase

@attribute Age {young,middle,old}

@attribute Income {low,medium,high}

@attribute Buys {yes,no}

@data

young,low,no

young,medium,yes

middle,medium,yes

old,high,yes

old,low,no

middle,high,yes

young,high,no

old,medium,yes

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

Choose **J48 -C 0.25 -M 2**

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation Folds **10**

☐ Percentage split % **66**

(Nom) Buys

Result list (right-click for options)

11:39:27 - trees.J48

Classifier output

```

=== Run information ===

Scheme:      weka.classifiers.trees.J48 -C 0.25 -M 2
Relation:    product_purchase
Instances:    8
Attributes:   3
              Age
              Income
              Buys
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

J48 pruned tree
-----

Income = low: no (2.0)
Income = medium: yes (3.0)
Income = high: yes (3.0/1.0)

Number of Leaves :    3

Size of the tree :    4

Time taken to build model: 0 seconds

```

Status

Problem evaluating classifier

 x 0

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

Choose **Logistic -R 1.0E-8 -M -1 -num-decimal-places 4**

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation Folds **10**

☐ Percentage split % **66**

(Nom) Buys

Result list (right-click for options)

11:39:27 - trees.J48

11:40:20 - functions.Logistic

Classifier output

```

Buys
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

Logistic Regression with ridge parameter of 1.0E-8
Coefficients...

Variable      Class
              yes
=====
Age=young      -27.1834
Age=middle     7.6504
Age=old        21.0631
Income=low     -47.6377
Income=medium  44.6584
Income=high    -6.5482
Intercept      13.362

Odds Ratios...

Variable      Class
              yes
=====
Age=young      0
Age=middle     2101.3903
Age=old        1404706923.4269
Income=low      0
Income=medium   2.482486790674901E19
Income=high     0.0014

Time taken to build model: 0.06 seconds

```

Status

Problem evaluating classifier

 x 0

EXPERIMENT 7:

Create the dataset using ARFF file format:

Transaction ID	Items
T1	Hot Dogs, Buns, Ketchup
T2	Hot Dogs, Buns
T3	Hot Dogs, Coke, Chips
T4	Chips, Coke
T5	Chips, Ketchup
T6	Hot Dogs, Coke, Chips

- Find the **frequent itemsets** and generate **association rules** on this. Assume that minimum support threshold ($s = 33.33\%$) and minimum confident threshold ($c = 60\%$).
- List the various rule generated by apriori and FP tree algorithm ,mention wheather accepted or rejected.

ARFF CODE:

```
@relation hotdogs_dataset
```

```
@attribute HotDogs {t,f}
```

```
@attribute Buns {t,f}
```

```
@attribute Ketchup {t,f}
```

```
@attribute Coke {t,f}
```

```
@attribute Chips {t,f}
```

```
@data
```

```
t,t,t,f,f
```

```
t,t,f,f,f
```

```
t,f,f,t,t
```

```
f,f,f,t,t
```

```
f,f,t,f,t
```

```
t,f,f,t,t
```

Weka Explorer

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator

Choose **Apriori** -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1

Start Stop

Result list (right-click for ...)

11:43:12 - Apriori

Associator output

```

=== Associator model (full training set) ===

Apriori
=====

Minimum support: 0.55 (3 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 9

Generated sets of large itemsets:

Size of set of large itemsets L(1): 6

Size of set of large itemsets L(2): 7

Size of set of large itemsets L(3): 4

Size of set of large itemsets L(4): 1

Best rules found:

1. Chips=t 4 ==> Buns=f 4 <conf:(1)> lift:(1.5) lev:(0.22) [1] conv:(1.33)
2. Buns=f 4 ==> Chips=t 4 <conf:(1)> lift:(1.5) lev:(0.22) [1] conv:(1.33)
3. Coke=t 3 ==> Buns=f 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
4. Coke=t 3 ==> Ketchup=f 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
5. Coke=t 3 ==> Chips=t 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
6. Ketchup=f Coke=t 3 ==> Buns=f 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
7. Buns=f Coke=t 3 ==> Ketchup=f 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
8. Buns=f Ketchup=f 3 ==> Coke=t 3 <conf:(1)> lift:(2) lev:(0.25) [1] conv:(1.5)
9. Coke=t 3 ==> Buns=f Ketchup=f 3 <conf:(1)> lift:(2) lev:(0.25) [1] conv:(1.5)
10. Ketchup=f Chips=t 3 ==> Buns=f 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)

```

Status OK Log x0

Weka Explorer

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator

Choose **FPGrowth** -P 2 -I -1 -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1

Start Stop

Result list (right-click for ...)

11:43:12 - Apriori

11:43:58 - FPGrowth

Associator output

```

=== Run information ===

Scheme:      weka.associations.FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1
Relation:    hotdogs_dataset
Instances:   6
Attributes:  5
             HotDogs
             Buns
             Ketchup
             Coke
             Chips

=== Associator model (full training set) ===

FPGrowth found 7 rules (displaying top 7)

1. [HotDogs=f]: 2 ==> [Buns=f]: 2 <conf:(1)> lift:(1.5) lev:(0.11) conv:(0.67)
2. [Chips=f]: 2 ==> [Coke=f]: 2 <conf:(1)> lift:(2) lev:(0.17) conv:(1)
3. [Ketchup=f, HotDogs=f]: 1 ==> [Buns=f]: 1 <conf:(1)> lift:(1.5) lev:(0.06) conv:(0.33)
4. [Ketchup=f, Coke=f]: 1 ==> [Chips=f]: 1 <conf:(1)> lift:(3) lev:(0.11) conv:(0.67)
5. [Ketchup=f, Chips=f]: 1 ==> [Coke=f]: 1 <conf:(1)> lift:(2) lev:(0.08) conv:(0.5)
6. [Buns=f, Coke=f]: 1 ==> [HotDogs=f]: 1 <conf:(1)> lift:(3) lev:(0.11) conv:(0.67)
7. [Coke=f, HotDogs=f]: 1 ==> [Buns=f]: 1 <conf:(1)> lift:(1.5) lev:(0.06) conv:(0.33)

```

EXPERIMENT 8:

Prediction of Categorical Data using Rule base classification and decision tree classification through WEKA using any datasets. Compare the accuracy using two algorithm and plot the

graph

@relation student_performance

@attribute StudyHours {low,medium,high}

@attribute Attendance {poor,average,good}

@attribute Result {pass,fail}

@data

low,poor,fail

low,average,fail

medium,average,pass

medium,good,pass

high,good,pass

high,average,pass

low,good,fail

medium,poor,fail

high,poor,pass

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

Choose J48 -C 0.25 -M 2

Test options

☐ Use training set

☐ Supplied test set Set...

☒ Cross-validation Folds 10

☐ Percentage split % 66

More options...

(Nom) Result

Start Stop

Result list (right-click for options)

11:47:54 - trees.J48

Classifier output

```
=== Run information ===

Scheme:      weka.classifiers.trees.J48 -C 0.25 -M 2
Relation:    student_performance
Instances:    9
Attributes:   3
              StudyHours
              Attendance
              Result
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

J48 pruned tree
-----

StudyHours = low: fail (3.0)
StudyHours = medium: pass (3.0/1.0)
StudyHours = high: pass (3.0)

Number of Leaves :    3

Size of the tree :    4

Time taken to build model: 0 seconds
```

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

Choose JRip -F 3 -N 2.0 -O 2 -S 1

Test options

☐ Use training set

☐ Supplied test set Set...

☒ Cross-validation Folds 10

☐ Percentage split % 66

More options...

(Nom) Result

Start Stop

Result list (right-click for options)

11:47:54 - trees.J48

11:48:35 - rules.JRip

Classifier output

```
=== Run information ===

Scheme:      weka.classifiers.rules.JRip -F 3 -N 2.0 -O 2 -S 1
Relation:    student_performance
Instances:    9
Attributes:   3
              StudyHours
              Attendance
              Result
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

JRIP rules:
=====

(StudyHours = low) => Result=fail (3.0/0.0)
=> Result=pass (6.0/1.0)

Number of Rules : 2

Time taken to build model: 0.01 seconds
```

